





DEPARTMENT OF AIML
AI PLANT DISEASE DETECTION

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CLASS AND SEC :3rd year &'A' sec

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Abstract

Plant diseases can reduce crop quality and yield. This project presents an image-based system that uses transfer learning to identify plant leaf diseases from uploaded photographs.

DATASET

PlantVillage dataset

38 leaf-condition classes

Multiple plant species

MODEL

MobileNetV2 pretrained on ImageNet

Feature extraction + classification

TensorFlow / Keras

APPLICATION

Flask web interface

Upload a leaf image

Display predicted class

The goal is to provide a simple decision-support prototype; predictions should be validated before real agricultural use.

Problem Statement

Early identification of plant diseases is important, but visual diagnosis can be difficult and time-consuming.

MANUAL INSPECTION

Requires experience and careful observation.

Symptoms may look similar across diseases.

DELAYED DETECTION

Late identification can allow infection to spread and may affect crop productivity.

ACCESSIBILITY

Growers may not always have immediate access to plant pathology expertise.

Problem to address: build an accessible image-classification prototype that predicts a likely leaf condition from an uploaded image.

Objectives

01

Prepare data

Organise leaf images into labelled classes and preprocess them consistently.

02

Train a model

Apply transfer learning with MobileNetV2 for multi-class classification.

03

Build an interface

Develop a Flask application for image upload and prediction.

04

Evaluate output

Assess model performance using held-out test data and suitable metrics.

Literature Review

Mohanty et al. (2016)

Demonstrated deep learning for image-based plant disease identification using the PlantVillage dataset.

Hughes & Salathé (2015)

Introduced the PlantVillage image dataset as a resource for automated plant disease classification research.

Howard et al. (2017)

Presented MobileNets, an efficient convolutional neural network family designed for mobile and resource-constrained applications.

Sandler et al. (2018)

Introduced MobileNetV2, using inverted residuals and linear bottlenecks for efficient visual recognition.

These studies motivate the use of labelled leaf images and lightweight transfer-learning architectures.

Research Gap

DATA CONDITIONS

Public datasets often contain clean, well-framed images. Real-world photos may have varied lighting, backgrounds and camera quality.

DEPLOYMENT

Many classification studies focus on model performance; a simple end-to-end upload-and-predict workflow is also needed.

GENERALISATION

High performance on a curated dataset does not guarantee reliable predictions on unseen field images.

Project focus: combine a transfer-learning classifier with a usable Flask prototype, while recognising dataset and field-generalisation limitations.

Existing System

TRADITIONAL APPROACH

- Manual visual inspection
- Expert consultation when available
- Diagnosis depends on experience
- Time-consuming for large numbers of plants

COMMON LIMITATIONS

- Subjective assessment
- Similar symptoms can be confused
- Diagnosis may be delayed
- Not always available at the point of need

Computer-vision approaches can support screening, but should not replace expert confirmation.

Proposed System

Input

User uploads a plant leaf image through the browser.

Preprocess

Resize image to 224 × 224 and apply the model's required scaling.

Classify

MobileNetV2-based classifier predicts one of the trained classes.

Display

Flask presents the predicted class to the user.

Expected outcome: a browser-based prototype that returns a predicted leaf-condition label from an uploaded image.

System Architecture

Architecture diagram supplied for the project

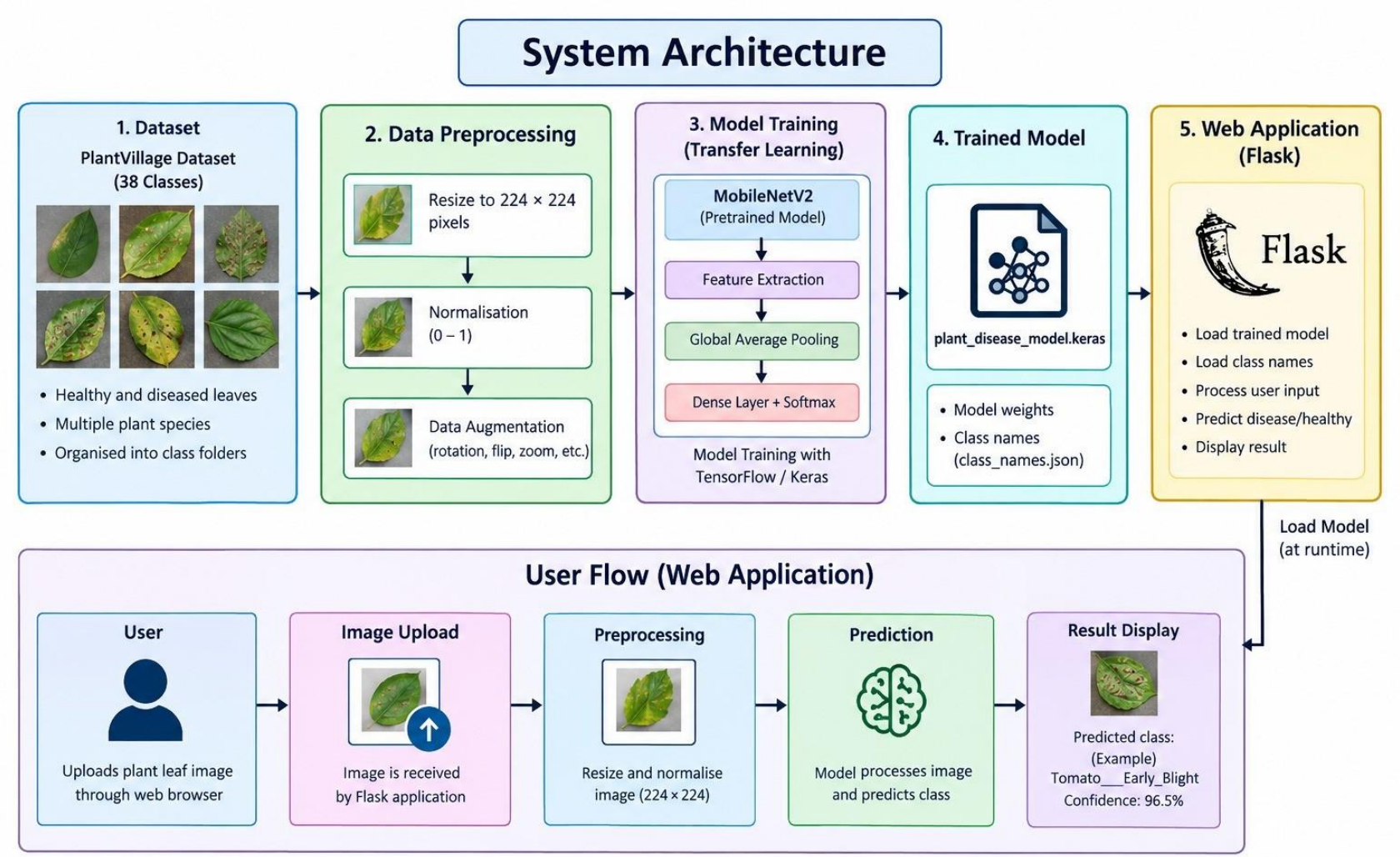


Figure 3.1 System Architecture of AI-Based Plant Disease Detection

Methodology

1. Dataset preparation

Use the PlantVillage folder structure with 38 labelled classes.

2. Preprocessing

Resize images to 224 × 224; normalise pixel values as required by the selected MobileNetV2 pipeline.

3. Augmentation

Apply training-time transformations such as rotation, horizontal flip and zoom.

4. Transfer learning

Load ImageNet-pretrained MobileNetV2; attach pooling and a Dense + Softmax classifier.

5. Training & evaluation

Train with labelled data; monitor validation performance and evaluate on a separate test set.

6. Inference

Load the saved .keras model and class_names.json in the Flask application.

Technology Stack

Python

Core programming language

TensorFlow / Keras

Model building, training and inference

MobileNetV2

ImageNet-pretrained transfer-learning backbone

Flask

Backend web framework and prediction route

HTML / CSS

Browser interface and result presentation

PlantVillage

Leaf image dataset with 38 classes

VS Code

Development environment

Pillow / NumPy

Image handling and numerical preprocessing

GANTT CHART — PROJECT TIMELINE

Project Duration: August 1, 2026 – October 31, 2026 • Dates retained from the reference PPT

Task / Activity	Days	AUGUST 2026	SEPTEMBER 2026	OCTOBER 2026
Requirement Analysis & Planning	7	Aug 1 – Aug 7		
Literature Survey & Feasibility Study	10	Aug 8 – Aug 17		
System Design (Architecture & Modules)	12	Aug 18 – Aug 29		
Dataset Collection & Preparation	14	Aug 25 – Sep 7		
Model Development (Prediction)	20		Aug 30 – Sep 18	
Budget Optimization Module	18		Sep 10 – Sep 27	
Dashboard & Cloud Integration	10		Sep 20 – Sep 29	
API / Service Integration	17		Sep 24 – Oct 10	
Testing, Optimization & Bug Fixing	12			Oct 11 – Oct 22
Final Evaluation, Documentation & Presentation	10			Oct 21 – Oct 30

Planning · Research · Design · Data · Development · AI/ Optimization · Integration · Testing · Finalization

Conclusion

The project combines deep learning and a web interface to support image-based plant leaf disease identification.

MODEL

MobileNetV2 transfer learning provides the classification backbone.

WORKFLOW

Preprocessing, prediction and result display are connected through Flask.

NEXT STEPS

Evaluate on independent field images, improve robustness and validate predictions with experts.

The system is a prototype and its reliability depends on the training data and evaluation results.

References

[1] S. P. Mohanty, D. P. Hughes, and M. Salathé, “Using Deep Learning for Image-Based Plant Disease Detection,” Frontiers in Plant Science, vol. 7, 2016.

[2] D. P. Hughes and M. Salathé, “An open access repository of images on plant health to enable the development of mobile disease diagnostics,” 2015.

[3] A. G. Howard et al., “MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications,” arXiv:1704.04861, 2017.

[4] M. Sandler et al., “MobileNetV2: Inverted Residuals and Linear Bottlenecks,” Proc. IEEE/CVF CVPR, 2018.

[5] TensorFlow, “Transfer learning and fine-tuning,” TensorFlow documentation.

[6] Flask, “Flask Documentation,” Pallets Projects.

Dataset: PlantVillage image dataset (38 classes), as used in this project.

THANK YOU

Questions & Discussion

AI-Based Plant Disease Detection Using Deep Learning and Transfer Learning